The spectral gap of a projection scan is at most its modified gap, which is at most the square root of n times the gap

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Editorial note. The counted statement is the author's labelled lemma environment at source0.tex lines 396-402; its proof environment opens at line 403, immediately after \end{lemma}, so the pairing is by adjacency and the retained proof range 403-433 is the whole environment.

Editorial note. This source is newly approved for this line and its preamble is built from the author's own package line, macro declarations and theorem-environment family (source lines 4-70); the heavy drawing, algorithm and font packages of the source are not retained because the retained material does not use them.

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Editorial scope. One original result of the pinned author manuscript, counted once: the two-sided comparison between the spectral gap, the modified gap and the coordinate gaps of a projection scan (author label lem:gibbs_comp).

Allowed standard prerequisites (editorial).

- Projection scan. A projection scan is a product of transition matrices, one for each coordinate, applied in a fixed order; P = P_n ... P_1 is the transition matrix of the systematic scan.. It is the transition matrix whose gaps are compared.
- Spectral gap and modified gap. The gap gamma of P is one minus the second largest eigenvalue in modulus, and the modified gap is the analogous quantity for the Dirichlet form of a single coordinate transition.. They are the two quantities appearing in the statement.

- Dirichlet form. For a reversible chain with stationary measure pi, the Dirichlet form of a function is the average of the squared differences weighted by the edge conductances.. It is the quantity used to compare the two gaps.

Lemma 0.1. *Let $\mathcal{X} = \mathcal{X}_1 \times \ldots \times \mathcal{X}_n$ be a finite product space and let $\mathsf{P} = \mathsf{P}_n \ldots \mathsf{P}_1$ denote the transition matrix of the systematic scan with identity permutation.*[1] *Then*

$$\gamma_{\mathsf{P}} \leq \widetilde{\gamma_{\mathsf{P}}} \leq \sqrt{2n\gamma_P}.$$

Proof. The first inequality holds for any P; indeed, for any f such that $\mathbb{E}_\pi[f] = 0$ and $\|f\|_\pi = 1$, the reverse triangle inequality implies that

$$\|(I - \mathsf{P})f\|_\pi \geq \|f\|_\pi - \|Pf\|_\pi \geq 1 - \|P\|_{\pi,\mathsf{op}}.$$

Since this holds for any such f, the first inequality follows.

The second inequality is a special property of Gibbs samplers. Since each P_i is an orthogonal projection in $L^2(\mathcal{X}, \pi)$, we will be able to precisely track the movement of f under each projection. Let f with $\mathbb{E}_\pi[f] = 0$ and $\|f\| = 1$ attain the maximum for $\|\mathsf{P}\|_{\pi,\mathsf{op}}$ so that

$$(1 - \gamma_{\mathsf{P}})^2 = \|\mathsf{P}f\|_\pi^2$$

Define $\gamma_0 = 0$ and for each $1 \leq i \leq n$, set $\mathsf{Q}_i = \mathsf{P}_i \ldots \mathsf{P}_1$ and γ_i via

$$1 - \gamma_i = \|\mathsf{Q}_i f\|_\pi^2.$$

Note that this sequence is nondecreasing since norms are nonincreasing under projections and that $\gamma_n = 2\gamma_{\mathsf{P}} - \gamma_{\mathsf{P}}^2 \leq 2\gamma_{\mathsf{P}}$. Moreover, the Pythagorean theorem implies

$$\|\mathsf{Q}_i f - \mathsf{Q}_{i-1} f\|_\pi^2 = \|\mathsf{Q}_{i-1} f\|_\pi^2 - \|\mathsf{Q}_i f\|_\pi^2 = \gamma_i - \gamma_{i-1}.$$

For this f, we thus have by the triangle inequality and Cauchy-Schwarz that

$$\begin{aligned}
\|(I - \mathsf{P})f\|_\pi^2 &= \|f - \mathsf{Q}_n f\|_\pi^2 \\
&= \left\| \sum_{i=1}^n (\mathsf{Q}_i f - \mathsf{Q} f) \right\|_\pi^2 \\
&\leq \left(\sum_{i=1}^n \|\mathsf{Q}_i f - \mathsf{Q}_{i-1} f\|_\pi \right)^2 \\
&\leq n \cdot \sum_{i=1}^n \|\mathsf{Q}_i f - \mathsf{Q}_{i-1} f\|_\pi^2 \\
&= n \cdot \sum_{i=1}^n (\gamma_i - \gamma_{i-1}) \\
&= n \cdot (\gamma_n - \gamma_0) \\
&\leq 2n\gamma_{\mathsf{P}}.
\end{aligned}$$

[1] Again, the actual Markov chain rerandomizes in the order $n, n-1, \ldots, 1, n, \ldots$ by our convention.

Taking roots completes the proof by the variational formula of the second smallest singular value of the Laplacian. □